

SREEHITH YACHAMANENI

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SUMMARY

Data Analyst with 2 years experience building **machine learning models, statistical experiments, and large-scale data pipelines** to extract insights from complex datasets. Skilled in **Python, SQL, predictive modeling, and feature engineering**, with hands-on experience developing ML systems for workforce analytics, business optimization. Passionate about applying machine learning and statistical analysis to solve real-world problems.

SKILLS

- Programming Languages: Python, R, SQL, C, C++, Golang
- Data Tools: Excel, MySQL, Tableau, IBM Cognos
- Data Skills: Data analysis, Machine Learning, Deep Learning, Data modeling, Data warehousing, Web scrapping, Data Wrangling, Data visualization, A/B Testing, Hypothesis Testing, Statistical Modeling

EDUCATION

Master of Computer Science

North Carolina State University Raleigh, North Carolina

GPA: 3.6/4.00

August 2023 - July 2025

WORK EXPERIENCE

Ecological Servants Project

Data Analysis and Research Volunteer

USA
November 2025 - Present

- Analyzed environmental impact and community outreach data using **Python, Pandas, and SQL** to track volunteer participation, waste reduction efforts, and sustainability program effectiveness across ecological service initiatives.
- Designed interactive dashboards and data visualizations in **Tableau** to present insights on recycling drives, conservation activities, and resource utilization, helping project coordinators make data-driven operational decisions.
- Cleaned, transformed, and validated multi-source environmental datasets, applying statistical analysis to identify participation trends and improve reporting accuracy for sustainability and community engagement programs.

Jio Platforms Limited

Data Analyst

Hyderabad-India
June 2022 - July 2023

- Analyzed large-scale telecom operational datasets using **Python and SQL** to identify usage patterns, operational bottlenecks, and key performance drivers across multiple business units.
- Developed automated **data preprocessing pipelines using Pandas and NumPy**, improving data quality and reliability by **15%** for analytics workflows.
- Optimized complex SQL queries and implemented indexing strategies on relational databases, reducing data retrieval latency by **25%** for reporting and analytics workflows.

PROJECTS

AI-Powered Elephant Range Extension News Intelligence System - [View Project on GitHub](#)

- Built an **AI-assisted data extraction pipeline** to process manually curated elephant-related news articles from Central India to support ecological research on **elephant range expansion patterns**.
- Integrated **Google Gemini API** to extract structured information such as **Date, location, conflict incidents, elephant deaths, and environmental context**.
- Developed scalable web scraping workflows using **Python, Requests, BeautifulSoup, and Selenium** to collect and process semi-structured web data while handling dynamic page content and inconsistent HTML structures.
- Designed structured datasets from extracted article content to enable **downstream spatial analysis, trend identification, and wildlife movement research**, improving the efficiency of manual research workflows

Vendor Performance Analysis - Retail Inventory and Sales using SQL and Python - [View Project on GitHub](#)

- Built a **SQL-based data pipeline** integrating multiple relational tables to analyze vendor performance across retail inventory and sales datasets, enabling large-scale supplier performance evaluation.
- Developed optimized SQL queries with **CTEs and aggregations** to generate vendor-level performance metrics, significantly reducing processing time for large transactional datasets.
- Conducted **exploratory data analysis (EDA) and hypothesis testing** using Python (Pandas, NumPy) to assess vendor profitability, pricing strategies, and inventory turnover patterns.
- Identified vendor concentration risk where **top 10 vendors accounted for 65.7% of total purchases**, highlighting potential supply chain vulnerabilities. Discovered **\$2.71M in unsold inventory** from low-performing vendors.

Taxi Fare Estimation and Tip Prediction – NYC Taxi and Limousine Commission - [View Project on GitHub](#)

- Built a **regression-based predictive model** to estimate NYC taxi fares using trip-level features such as distance, duration, passenger count, and pickup/drop-off locations, improving fare estimation accuracy through feature engineering and model tuning.
- Developed a **Random Forest classification model** to predict the probability of passengers leaving tips, leveraging behavioral and transactional features extracted from the NYC Taxi dataset.
- Performed comprehensive **data preprocessing and feature engineering** on large trip datasets, including handling missing values, outlier detection, and transformation of temporal and geospatial features.
- Evaluated model performance using **cross-validation and classification metrics (precision, recall, F1-score)** to ensure robustness and minimize prediction bias.